JCLAP · MIDI NOTE EFFECT · MAX FOR LIVE

Polar XY

Take control of the Orb in Omnisphere 3. User Manual



VERSION	FORMAT	PLATFORMS	EDITION
0.19.3	CLAP · VST3 · Max for Live	macOS · Windows · Linux	Draft · November 2026

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About this manual

Chapters 2 and 3 get you installed and steering the Orb in a few minutes. Chapter 4 tours the panel, and Chapter 5 explains how an X/Y position becomes an angle and a radius, which is the quickest way to understand what every control does. Chapters 6 to 9 cover the controls section by section. Chapters 10 and 11 are for **Ableton Live**: the Max for Live device, its setup, and MIDI-learning the Orb in Omnisphere 3. Chapter 12 has ready-made setups for different controllers. The appendix lists every parameter with its range and default, plus specifications and troubleshooting.

Throughout the manual, control names are set in **bold** and panel labels as they appear on screen, such as **INERTIA MODE**. Where the Max for Live device labels a control differently, the device's label follows in brackets, such as **Smoothing** (**Smooth**).

M4L Also on the Max for Live device, under the label shown.

CHAPTER 01

Welcome to Polar XY

A small MIDI effect that turns any X/Y controller into a steering wheel for the Orb in Omnisphere 3.

The Orb in Omnisphere 3 is a circle. Where its dot sits is described the way a radar would: an **angle** round the circle and a **radius** out from the centre. Omnisphere lets you MIDI-learn each of those to a CC. An X/Y pad, a joystick or a touch controller doesn't speak that language: it sends one CC for left and right and another for up and down. Learn those straight onto the Orb and pushing the stick right turns the Orb instead of moving it right.

Polar XY sits in front of Omnisphere and does the translation. It listens for your controller's two CCs, works out where on the circle that point is, and sends the angle and the radius on two CCs of their own, ready to MIDI-learn onto the Orb. Push right and the Orb goes right. Circle the stick and the Orb circles. Everything else on the MIDI channel (notes, pitch bend, other CCs) passes straight through.

Around that conversion are the controls that make it feel good: **Smoothing** and **Inertia** give the Orb weight so it glides, swings and coasts; a **Touch Filter** cleans up the skitter a resistive pad makes when a finger presses and lifts; and **14-bit output** takes the stairs out of a slow sweep.

At a glance

- **X/Y in, angle and radius out**, on any CCs you choose, on every MIDI channel independently.
- **Angle Offset and Direction** line the controller up with the Orb, however the Orb is rotated or mirrored.
- **Clamp to circle or Fill square**, so a square pad's corners do what you expect.
- **Motion**: Smoothing up to 2 s, Inertia with overshoot and coasting, Straight or Around paths, and Pull, Coast and Pad modes.
- **Touch Filter** for resistive pads, with an optional touch CC so it knows exactly when a finger is down.
- **Center zone** that steadies the angle around the middle.
- **MIDI learn buttons** that send one CC alone, so Omnisphere learns the right one.
- **Three formats**: CLAP, VST3, and a Max for Live device for Ableton Live.

> **WHICH VERSION SHOULD I USE?**
> In **Bitwig Studio**, REAPER or any host that loads CLAP note effects, use the CLAP. In **Ableton Live**, use the Max for Live device (Chapter 10): Live doesn't load CLAPs, and won't put a VST3 in front of an instrument on the same track. The VST3 is for other hosts that can route a plug-in's MIDI output on to the next device.

CHAPTER 02

Installation

Polar XY ships in the JClap bundle as a CLAP, a VST3 and a Max for Live device.

Requirements

CLAP	CLAP 1.2 note effect. Your host must support CLAP note effects, for example Bitwig Studio or REAPER.
VST3	Any VST3 host that can pass a plug-in's MIDI output on to the next device. It shows up as an instrument, since VST3 has no MIDI-effect category.
Max for Live	Ableton Live Suite, or Live Standard with Max for Live. See Chapter 10.
macOS	Universal binary: macOS 10.15 or later on Intel, 11 or later on Apple Silicon.
Windows	64-bit Windows.
Linux	64-bit, X11 (CLAP and VST3).

Installing the plug-in

01 **Download**

Grab the zip from the latest release at `github.com/jeffmhopkins/Jclap/releases/latest`. `jclap-macOS.zip` and `jclap-Windows.zip` hold every JClap plug-in; `jclap-PolarXY-0.19.3.zip` holds only Polar XY, with a folder per system.

02 **Copy the plug-in**

Copy `PolarXY.clap` into your CLAP folder, and `PolarXY.vst3` into your VST3 folder if you want it:

	CLAP	VST3
macOS	`~/Library/Audio/Plug-Ins/CLAP`	`~/Library/Audio/Plug-Ins/VST3`
Windows	`C:\Program Files\Common Files\CLAP`	`C:\Program Files\Common Files\VST3`
Linux	`~/.clap`	`~/.vst3`

03 **Rescan**

Restart your host or rescan plug-ins. In Bitwig Studio this is under *Settings → Locations*. Polar XY appears as a note effect.

MACOS: BROWSER DOWNLOADS ARE QUARANTINED

The plug-ins are ad-hoc signed but not notarized, so macOS quarantines files downloaded through a web browser and some hosts (Bitwig among them) silently skip them. Clear the flag once after copying:

```
xattr -dr com.apple.quarantine ~/Library/Audio/Plug-Ins/CLAP/*.clap ~/Library/Audio/Plug-Ins/VST3/PolarXY.vst3
```

Where it goes

Polar XY works on MIDI, not audio, so it goes on the **Omnisphere track, in front of Omnisphere**. Your controller's MIDI reaches Polar XY first; the two X/Y CCs are swallowed and turned into the angle and radius CCs, and everything else carries on to Omnisphere unchanged. No extra tracks or MIDI routing are needed.

The Max for Live device is installed differently: see *Chapter 10*.

CHAPTER 03

Quick Start

From an empty slot to steering the Orb, in Bitwig Studio. For Ableton Live, skip to Chapter 10.

01 **Load it in front of Omnisphere**
On the Omnisphere track, put **Polar XY** before Omnisphere in the device chain.

02 **Tell it your controller's CCs**
Set **X** and **Y** under XY IN to the CCs your controller sends (16 and 17 by default; Chapter 12 lists the defaults for some popular controllers). Move the controller: the dot in the **XY IN** view follows it, and the orb in **POLAR OUT** follows that, with the angle and radius it is sending shown underneath.

03 **Pick the output CCs**
ANGLE and **RADIUS** under POLAR OUT are the CCs that go to Omnisphere (18 and 19 by default). Any CCs your controller and Omnisphere aren't already using will do.

04 **Learn the angle**
Open Omnisphere's **ORB** page and switch the Orb on. Right-click the Orb, check the menu says the Orb is set for *Angle* (if it offers *Switch to Angle*, choose it first), and choose **MIDI CC Learn**. Then press **ANGLE** under MIDI LEARN on Polar XY. It sends the angle CC alone, once, so Omnisphere learns exactly that CC.

05 **Learn the radius**
Right-click the Orb, choose **Switch to Radius**, right-click again, choose **MIDI CC Learn**, and press **RADIUS** on Polar XY.

06 **Line it up**
Move the controller. If the Orb's dot lands rotated from where you are pointing, turn **OFFSET**. If it moves the opposite way round, flip **DIRECTION**.

07 **Give it some weight**
Raise **SMOOTHING** to about 150 ms for a glide, or **INERTIA** to 40% to let the Orb swing past and settle. Click **SAVE** in the preset bar to keep the setup.

> TIP
> Every control has a tooltip. Hover over anything on the panel for a one-line reminder of what it does.

CHAPTER 04

The Panel

Input on the left, output on the right, and the settings that shape the journey between them along the bottom.



1. **Preset bar.** Step with the arrows, click the name for the menu, star favourites, **SAVE** your own.
2. **XY IN view.** The last position received. Click or drag to play the Orb from here.
3. **Input CCs.** X, Y and the optional Touch CC from your controller.
4. **PASS INPUT CCS.** Also send the X/Y CCs on: Off, Raw or Filtered.
5. **Polar Out switch.** Off sends no angle or radius at all.
6. **POLAR OUT view.** Where the output is, and where it is heading.
7. **Output CCs.** Angle, Radius and the optional Touch Out CC.
8. **MIDI LEARN.** Send one output CC alone, for Omnisphere to learn.
9. **SETTINGS / TOUCH tabs.** Mapping, motion and resolution; Touch Filter tuning and Center.

The two views

XY IN shows the last X/Y position received, with its channel, and **POLAR OUT** the angle and radius sent for that channel. Angle 0 is marked by the tick on the right of the circle. The numbers under each view read the same thing out: *X*, *Y* and *CH* on the left, *ANGLE* and *RADIUS* on the right (a dash until both axes have arrived). Comparing the two views shows exactly what Offset and Direction are doing.

Both views are small night skies. The input dot and the output orb leave a trail of ink that drifts and fades behind them, so you can see the path a glide took (straight across, or round the circle) and Inertia swinging it past the target and back. The orb is drawn where the output really is, between the CC steps. While Smoothing or Inertia hold it back, a faint band ties it to a blue ring showing where the controller is pointing it.

Playing from the panel

Click or drag in **XY IN** and that channel goes there just as if the controller had sent it, gliding with Smoothing and Inertia like any other move (and, with Pass Input CCs on, sending the X/Y CCs on too). The pad isn't rounded to CC steps, and its moves always get at least 40 ms of smoothing, so the Orb follows the mouse evenly. While the mouse is down it has that channel, and the controller's X/Y on it are ignored; let go and the controller takes over with its next move. Double-click to put it back in the centre. Before any CC has arrived the pad drives channel 1.

Working with the controls

Every knob, switch and menu is a host parameter, so everything can be automated. Knobs work the same everywhere on the panel:

ACTION	ON A KNOB
Drag up / down	Change the value. A popup shows the value while you drag.
Shift + drag	Fine adjustment.
Scroll wheel	Step the value up or down.
Right-click	Menu with *Reset to default* and *Enter value…* to type an exact number, such as a CC.

Presets

The preset bar in the header steps through presets with its arrows, lists them when you click the name, and stars favourites. **SAVE** stores the current settings as your own preset; it appears under *User* in the menu. The factory bank has two: **Init**, with every setting at its default, and **Geren**, tuned for a resistive pad (Chapter 12). Settings are also saved with your project.

macOS	`~/Library/Application Support/PolarXY/Presets`
Windows	`%APPDATA%\PolarXY\Presets`
Linux	`~/.config/PolarXY/Presets`

Panel size

The panel is 900 × 738 at 100%. Right-click any empty part of the panel and choose *UI Size* to scale it to 75%, 100%, 125%, 150% or 200%. The same menu has *Check for Updates*.

CHAPTER 05

How Polar XY Works

The same point, described two ways: how far across and how far up, or which way and how far out.

From X/Y to angle and radius

Your controller sends a position as two numbers from 0 to 127, with 64 in the middle: X grows to the right and Y grows upward. Polar XY treats 64, 64 as the centre of the Orb. The **radius** is how far the point is from the centre: 0 there, 127 at the edge. The **angle** is which way the point lies, starting from straight right and turning counter-clockwise, in 127 steps for a full turn.

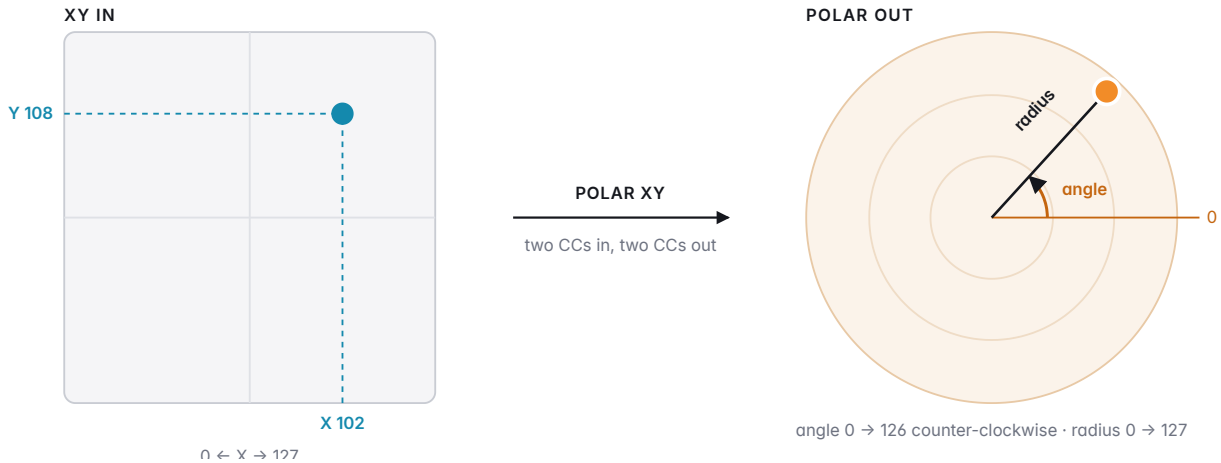


Fig. 5.1 The same point as X/Y and as angle and radius. The angle starts at the right (the tick on the POLAR OUT view) and turns counter-clockwise by default.

The details that matter

- **Angle steps.** One full turn is 127 steps, so the angle runs 0 to 126; 127 would be a full turn, the same place as 0, so the angle wraps from 126 straight to 0.
- **The dead centre.** Right at the middle (within about two CC steps) a point has no direction, and a resting controller's jitter would swing the angle wildly. So there Polar XY holds the last angle and sends radius 0. Outside that, the radius starts at 1.
- **The corners.** A square pad's corners are further from the centre than its edges, about 1.41 times. By default the radius stops at the circle; **Radius Mode** can instead make the whole square edge full radius (Chapter 6).
- **One move at a time.** X and Y arriving together are one move: only the final point goes out, never a point made of the new X and the old Y.
- **Only changes.** An output CC is sent only when its value changes.

- **Every channel is separate.** Each of the sixteen MIDI channels has its own position and its own motion. The output goes out on the channel the input came in on.

What passes through

Everything that isn't one of the input CCs passes straight through: notes, pitch bend, aftertouch, other CCs, clock. The input CCs themselves are swallowed, unless **Pass Input CCs** sends them on as well (Chapter 8). If X and Y In are set to the same CC, or Angle and Radius Out are, nothing is converted and the input passes through; the panel shows a warning.

> **WHY THE ORB MIGHT LOOK ROTATED**
> Spectrasonics doesn't document where the Orb's angle 0 sits or which way it turns, so Polar XY makes both settings: **Angle Offset** and **Direction**. If the Orb's dot lands rotated or mirrored from your controller, adjust those (Chapter 6).

CHAPTER 06

Mapping and CCs

Which CCs come in and go out, and how the controller's square lines up with the Orb's circle.

XY IN

Input CCs

CONTROL	RANGE	DEFAULT	WHAT IT DOES
X M4L · X CC	CC 0–127	CC 16	The CC your controller sends for left and right.
Y M4L · Y CC	CC 0–127	CC 17	The CC your controller sends for up and down.
TOUCH M4L · Touch	Off, CC 0–127	Off	The CC a touch pad sends while a finger is down. Optional; see Chapter 8. On the Max for Live device, −1 is Off.

Input takes **14-bit** CCs too, with nothing to set. When an input CC is below 32, a fine value on that CC + 32 (sent after the main one, as 14-bit controllers do) places the axis between the CC steps. A 7-bit controller never sends them, so nothing changes for one.

POLAR OUT

Output CCs

CONTROL	RANGE	DEFAULT	WHAT IT DOES
ANGLE M4L · Angle	CC 0–127	CC 18	The angle goes out on this CC, for the Orb's angle.
RADIUS M4L · Radius	CC 0–127	CC 19	The radius goes out on this CC, for the Orb's radius.
TOUCH M4L · Touch	Off, CC 0–127	Off	Sends the touch on, as 127 and 0. Leave it Off for Omnisphere.
OFF / ON M4L · POLAR OUT	Off, On	On	The switch in the POLAR OUT title. Off sends no angle or radius at all (Chapter 8).

SETTINGS

Lining up with the Orb

CONTROL	RANGE	DEFAULT	WHAT IT DOES
OFFSET M4L · Offset	0–360°	0°	Rotates the angle. Turn it until the Orb's dot sits on the same side as your controller.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
DIRECTION M4L · Direction	`CCW, CW`	`CCW`	Which way the angle counts. Flip it to CW if the Orb goes round the opposite way to your controller.
RADIUS MODE M4L · Radius	`Circle, Square`	`Circle`	*Circle* (Clamp to circle) stops the radius at the edge of the circle, so the corners are no further out than the edges. *Square* (Fill square) makes the whole edge of the square full radius.

> **CIRCLE OR SQUARE?**
> Joysticks move in a circle, so *Circle* fits them. On a square pad, *Circle* means the edges reach full radius only at their middles; *Square* lets you reach full radius by pushing to any edge, at the cost of the corners and edges feeling the same.

MIDI LEARN

Sending one CC alone

MIDI-learning an Orb control is awkward with an X/Y controller, because a controller never moves just one CC: Omnisphere learns whichever one gets there first. The **ANGLE**, **RADIUS** and **TOUCH** buttons under MIDI LEARN each send that output's CC alone, once, at the value last sent on it (64 before anything has been sent; TOUCH as it stands, or released if nothing has been sent), so the learn picks up exactly that CC. At 14-bit, only the main (MSB) CC is sent, which is the one to learn. TOUCH needs a Touch Out CC set.

On the Max for Live device these are the **SEND (MIDI LEARN)** buttons. Typing a number into the device's Angle, Radius or Touch Out box and pressing Enter sends the same way, even when the number is the one already there (Chapter 10).

CHAPTER 07

Motion

By default the Orb jumps to wherever the controller is. Motion makes it travel there instead, with as much weight as you like.

With **Smoothing** or **Inertia** up, Polar XY sends a stream of in-between CCs as the Orb travels (one step every 32 samples, or every 4 ms on the Max for Live device, and only when a value changes) until it arrives. Each MIDI channel moves on its own. The first position on a channel is jumped to, as there's nowhere to glide from.

SETTINGS

Smoothing and Inertia

CONTROL	RANGE	DEFAULT	WHAT IT DOES
SMOOTHING M4L · Smooth	0 ms – 2 s	0 ms	Roughly how long the Orb takes to glide to the controller's position.
INERTIA M4L · Inertia	0–100%	0%	Gives the Orb weight. The heavier it is, the slower it gets going and the further it swings past the target before settling. At 100% it swings for about 5 s; the low end of the knob stays light.

With Inertia up, the Orb doesn't stop when the controller does (or when you let go of the pad). It carries on the way the controller was going, slowing down over a couple of seconds at 100%, and bounces off the edge. It settles wherever it runs out of speed, and the controller's next move picks it up from there. A jump from rest has no speed to carry, so it settles on the spot.

SETTINGS

Path



Fig. 7.1 The same glide with Path on Straight and on Around. Straight crosses the pad through the middle; Around turns the angle and changes the radius separately, so it goes round the circle the short way.

SETTING	WHAT MOVES
Straight	Glides across the pad in X/Y, so going from one side to the other passes through the centre.
Around	Moves the angle and radius on their own, so the Orb travels round the circle. The angle keeps counting past a full turn, so stirring the controller with some Inertia sets the Orb whirling, and it carries on past where you stop. A radius that swings or coasts in past the centre comes out the other side rather than waiting there.

SETTINGS

Inertia Mode

INERTIA MODE (Inertia mode) picks how the Orb follows the controller once it has weight.

MODE	HOW IT FOLLOWS
Pull	The default. The Orb is drawn to where the controller is, swinging past and settling as described above.
Coast	Only the controller's *movement* moves the Orb. While the controller moves, the Orb picks up its speed and direction (slower the heavier it is). Once the controller holds still it has no say at all: the Orb coasts on, slows to a stop and stays there. Where the controller is doesn't matter, so the Orb can end up anywhere.
Pad	Treats the controller like a laptop trackpad, for a touch X/Y controller used as a chaos pad. Touching down doesn't move the Orb, wherever the finger lands. Dragging moves it by as much as the finger moves, from wherever it is. Letting go mid-drag flings it on, gliding and bouncing off the walls, with Inertia setting how far (at 0% it stops where you let go). Touching again catches it. With Path on Around, dragging across turns the Orb (a full sweep of the pad is a full turn) and dragging up and down changes the radius.

> **PADS THAT SPRING BACK**
> In Pad mode, a controller that jumps back to the centre when released is fine: any big jump counts as a finger landing or lifting, not as a drag.

CHAPTER 08

Touch Filter and Center

A resistive pad lies for a few milliseconds every time a finger lands or lifts. Touch Filter finds those moments and throws them away.

A resistive X/Y pad reads its position through the finger's contact. For a few milliseconds while the finger presses down, and again while it lifts off, the contact is light and the readings are wrong: they skitter toward an edge or the centre, and the Orb jumps there and back. Smoothing hides that by smearing it into every move. **Touch Filter** instead finds the edges of each touch and drops them, leaving the rest of the movement exactly as it came.

SETTINGS

Touch Filter

TOUCH FILTER (**Filter ms**) is a time: pick 10, 20, 40, 50 or 60 ms from its menu, or *Custom* for any time up to 100 ms. Off (0) passes everything as it comes. Use the shortest time that stops the skitter. It costs that much latency (plus the press at the start of each touch), but it never smears a move. The panel's own XY pad doesn't go through it.

- X and Y that arrive within 3 ms of each other are one reading, so no point made of the new X and the old Y goes out.
- A reading that jumps further than the **Jump Limit**, or lands somewhere new after a 50 ms pause, starts a touch. For the first stretch of it, the press, nothing goes out. Then the latest reading goes out if the last two **Agree**, and everything after flows on. A touch that never gets two readings, such as a single blip as the finger lifts, never goes out.
- Everything is held back by the Touch Filter time, so when a jump starts a new touch, the readings from just before it (the last touch lifting off) are thrown away. The last few milliseconds of a drag go with them.

TOUCH TAB

Tuning

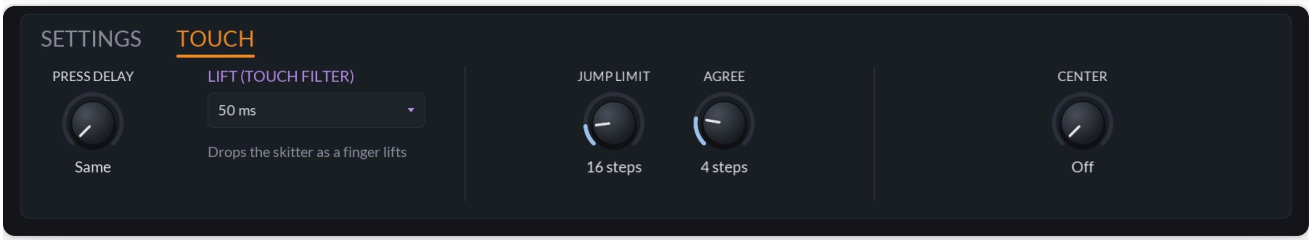


Fig. 8.1 The TOUCH tab, with Touch Filter at 50 ms. Touch Filter appears here too, as **LIFT (TOUCH FILTER)**.

The **TOUCH** tab, next to SETTINGS at the bottom of the panel, tunes the filter. On the Max for Live device these are in the **TOUCH** section.

CONTROL	RANGE	DEFAULT	WHAT IT DOES
PRESS DELAY M4L · Press delay	Same, 0–100 ms	Same	How long a new touch's press is held back. *Same* follows Touch Filter. On the device, −1 is Same.
JUMP LIMIT M4L · Jump	8–64 steps	16 steps	How far one reading may move before it has to wait for a second to agree. It's measured from the last reading and from where the finger was heading (its speed and direction carried on), whichever is nearer, so a fast drag that keeps its course passes, and so does a sharp turn back.
AGREE M4L · Agree	1–16 steps	4 steps	How close two readings must be to agree.
CENTER M4L · Center	Off, 0–50%	Off	A zone around the middle that steadies the angle (below).

TOUCH TAB

Center

CENTER is a zone around the middle, as a share of the radius. Inside it:

- **The angle holds still through small wobbles.** Near the centre, a wobble of a CC step or two swings the angle a long way, and across the centre it flips it half a turn. In the zone, the angle is read from a point the position drags along on a short string (4 CC steps): a wobble shorter than that doesn't move it, and a real move pulls it along at once. It goes by distance, not time, so a move through the centre is never held up. The string shrinks to nothing toward the zone's edge.
- **With Path on Around, motion goes straight.** Around would swing a glide across the middle round the centre; in the zone it fades to Straight, so the Orb cuts through the middle and orbits further out.

The radius always goes out exactly. Try 15 to 25%.

XY IN / POLAR OUT

The touch CC

Many pads also send a CC that is on (64 and up) while a finger is down and off when it lifts. Set **TOUCH** under XY IN to it, and Touch Filter knows the edges of each touch for certain instead of guessing them from jumps:

- While the touch is off, X and Y are ignored.
- When it goes on, the press is held back as above, then the latest reading goes out (one is enough, as the finger is known to be down). A jump later in the touch still waits for a second reading to agree, and a pause no longer counts as a new touch.
- When it goes off, everything still held back is thrown away: the last Touch Filter time before the lift. So a lift-off is dropped even when it slides away gently with no jump to give it away. The touch CC usually arrives a reading or more after the skitter starts, so use a time at least that long (40 to 50 ms for a pad that sends 60 times a second).

Until the first touch CC arrives, Touch Filter works as before, so a pad that doesn't send one loses nothing. The touch CC needs Touch Filter on; with it off, the touch CC only goes on to Touch Out.

TOUCH under POLAR OUT sends the touch on, as 127 and 0, in step with the filtered motion: on just before a touch's first position goes out, off once it lifts. Holding the mouse down on the panel's XY pad counts as a touch too.

XY IN

Passing the CCs on

PASS INPUT CCS (Pass) sends the X/Y CCs (and the touch CC) on as well as converting them.

SETTING	WHAT GOES ON
Off	The default. The input CCs are swallowed; only the angle and radius go out.
Raw	The input CCs go on the moment they arrive, exactly as the controller sent them, skitter and all.
Filtered	The input CCs go on as Touch Filter lets them through: the same CCs and values, a window later, with each touch's press and lift-off left out and the touch CC in step. With Touch Filter off, this is the same as Raw.

With **Polar Out** switched off, no angle or radius goes out at all, so Polar XY can be used purely to clean up a resistive pad's X/Y for something else.

CHAPTER 09

14-bit Output

Finer steps for slow sweeps, if the receiving end can use them.

A 7-bit CC gives the angle 127 steps round the circle (about 2.8° each) and the radius 127 steps, so a slow sweep can move in audible stairs. **RESOLUTION** (Resolution) set to **14-bit** sends each value as a standard 14-bit CC pair: the top 7 bits on the output CC, then the bottom 7 bits on that CC + 32. With the defaults, 18 and 19, the fine values also go out on 50 and 51. That is 16,383 steps each.

- **It needs both output CCs below 32.** On higher ones the output stays 7-bit, and the panel (or the device's status line) says so.
- Whether the receiving end reads the pair as one fine value is up to it. One that doesn't simply sees the main CC move as before, plus two extra CCs.

> **WITH OMNISPHERE**
> Start at **7-bit**, especially while MIDI-learning. If Omnisphere won't learn the Orb with 14-bit on, switch to 7-bit and learn again (Chapter 11).

CHAPTER 10

Using Polar XY with Ableton Live

In Live, Polar XY is a Max for Live MIDI effect that sits on the same track as Omnisphere 3.

Polar XY works best when it can sit on the same track as Omnisphere 3. Live doesn't load CLAPs, and won't put a VST3 in front of an instrument on the same track, so for Live there is a **Max for Live** version: `Polar XY.amxd`. It needs Ableton Live Suite, or Live Standard with Max for Live installed.

It is the same conversion and motion as the plug-in, with the same settings, each a Live parameter you can automate or map. In place of the plug-in's night-sky views, a status line shows the last channel's X/Y in and angle and radius out.

> **NOTE**
> This chapter and the next were prepared with Polar XY 0.19.3 and Ableton Live Suite 12.4.16.

The Max for Live device

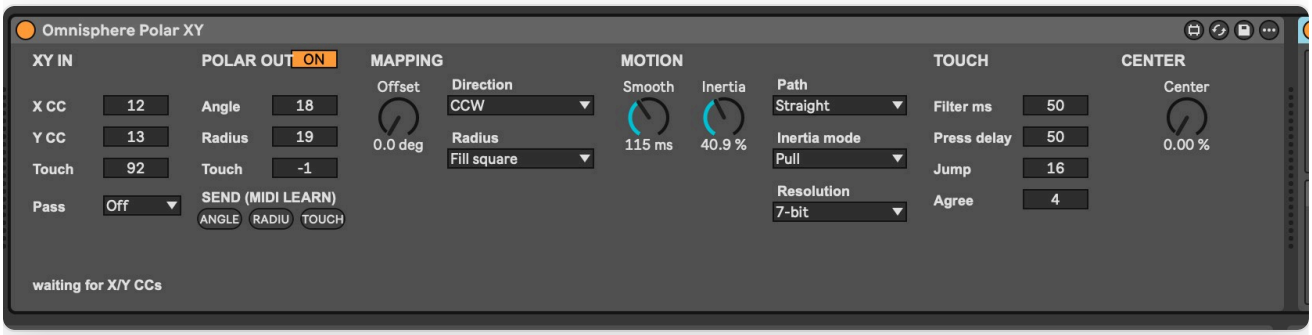


Fig. 10.1 The Polar XY device in Live, here saved as an *Omnisphere Polar XY* preset for a pad sending X on CC 12, Y on CC 13 and touch on CC 92.

The device has six sections, left to right:

SECTION	WHAT IT HOLDS	CHAPTER
XY IN	The Cartesian CC data coming from your X/Y controller: **X CC**, **Y CC**, **Touch** (−1 is off) and **Pass** (Off, Raw or Filtered).	6, 8
POLAR OUT	The CCs carrying the converted polar output sent to Omnisphere: **Angle**, **Radius**, **Touch** (−1 is off), the **OFF / ON** switch in the title, and the three **SEND (MIDI LEARN)** buttons.	6
MAPPING	How the Cartesian data maps onto Omnisphere's polar coordinates: **Offset**, **Direction** and **Radius** mode.	6
MOTION	How the Orb responds and moves: **Smooth**, **Inertia**, **Path**, **Inertia mode** and **Resolution**.	7, 9

SECTION	WHAT IT HOLDS	CHAPTER
TOUCH	Tuning for a jittery X/Y pad: **Filter ms** (0 is off), **Press delay** (−1 is the same as Filter ms), **Jump** and **Agree**.	8
CENTER	The dead-zone behaviour around the origin: **Center**.	8

The status line

Along the bottom of the device, the status line shows what Polar XY is doing, and ends with the device's version, so you can see which copy Live loaded.

IT SHOWS	MEANING
`waiting for X/Y CCs`	Nothing has arrived on the X or Y CC yet. Check the X CC and Y CC numbers, and that the controller's MIDI reaches the track.
`ch 1 in 101,91 out 35 / 90 v0.19.3`	The last channel that moved, its X,Y in, and the angle / radius sent. A dash means that value hasn't arrived or been sent yet; `out off` means Polar Out is switched off.
`angle: sent CC 18 = 64 on ch 1`	A SEND button (or Enter in an Out box) just sent that CC. Shown for a few seconds.
`(14-bit needs out CCs < 32)`	Resolution is 14-bit but an output CC is 32 or above, so the output stays 7-bit.
`… are the same CC: passing through`	X and Y In, or Angle and Radius Out, are set to the same CC, so nothing is converted.

Installation and initial setup

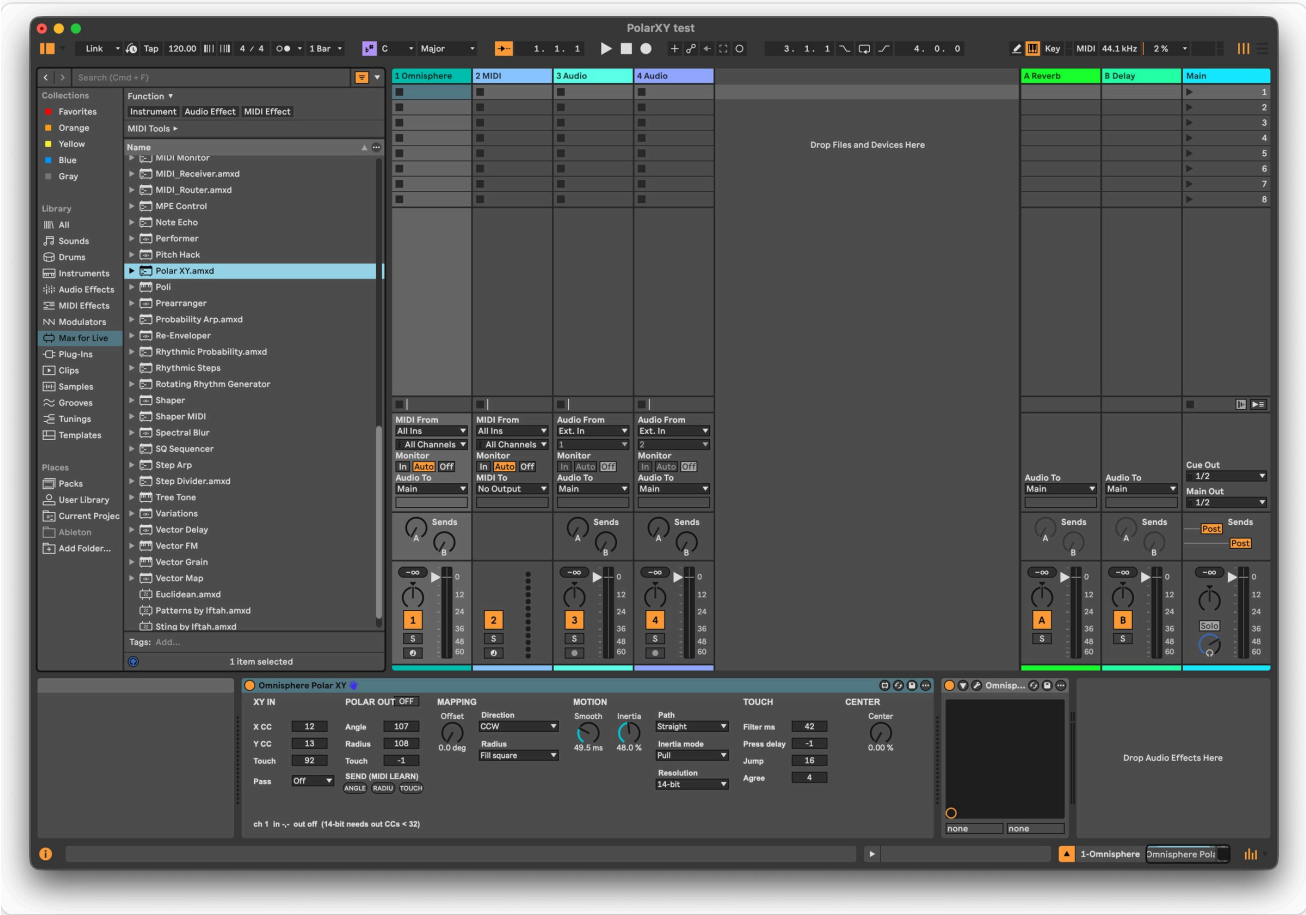


Fig. 10.2 Polar XY.amxd in Live's browser under *Max for Live*, and the device on the Omnisphere track.

01 Make a folder for it

In your Ableton Live User Library, create a folder for your Max MIDI effects, for example *MAX MIDI FX*. On a Mac the User Library is at `~/Music/Ableton/User Library/`.

02 Unzip the device

From the *Max for Live* folder of the release zip, unzip `Polar XY.amxd` and `polarxy.js` into the new folder. Keep the two files together: the device runs the script beside it.

03 Load it

Open Ableton Live and choose *Max for Live* in the browser. `Polar XY.amxd` is listed there. Drag it onto an available MIDI track. It loads with the default settings.

04 Check Pass and Polar Out

Make sure **Pass** under XY IN is *Off* (the default), so only the angle and radius reach Omnisphere, and that the switch beside **POLAR OUT** reads *ON*.

05 **Enter your controller's CCs**
In the **X CC** and **Y CC** boxes, enter the CC numbers your X/Y pad sends. Chapter 12 lists the defaults for some popular controllers.

06 **Set the angle CC**
Click the **Angle** value, type the CC you want for the angle and press Enter.

07 **Set the radius CC**
Click the **Radius** value, type the CC you want for the radius and press Enter.

08 **Leave Touch off**
For Omnisphere, the **Touch** value under POLAR OUT can be left at −1 (off).

09 **Set the resolution**
Under the MOTION heading, set **Resolution** to *7-bit*.

10 **Save a preset**
Click the small disk icon in the upper right of the device's title bar and save the settings as a preset, for example *Omnisphere Polar XY*.

That completes the basic setup of the Max for Live version. The next chapter learns the Orb in Omnisphere.

> **ONE SELF-CONTAINED FILE**
> To make the device a single file that doesn't need `polarxy.js` beside it, open it in the Max editor (the edit button in its title bar), press **Freeze Device**, and save.

CHAPTER 11

Configuring Omnisphere

Teaching the Orb which CCs to follow. Set up is easy, but tedious, and you only do it once.

The Max for Live version of Polar XY is a MIDI effect in Ableton Live, so it can be placed on the same MIDI track as Omnisphere 3. Setting the Orb's control CCs means a **MIDI learn**, which can be tricky, because an X/Y controller never sends just one CC. That is why Polar XY has dedicated buttons for the learn: each sends one CC alone.

Learning the Orb

01 **Start from your Polar XY track**
Add Polar XY to a MIDI track. The track from *Installation and initial setup* (Chapter 10) will do.

02 **Add Omnisphere**
Add Omnisphere 3 to the same track, after Polar XY, and open its plug-in window.

03 **Load a patch that uses the Orb**
For example *Clean Air at Last* from the *Ambient Dreams* directory.

04 **Keep both in view**
Move the Omnisphere window so it doesn't cover the Polar XY device.

05 **Switch the Orb on**
Go to the **ORB** page and click the Orb's power button.



Fig. 11.1 Omnisphere 3's ORB page with the Orb switched on, on the *Clean Air at Last* patch.

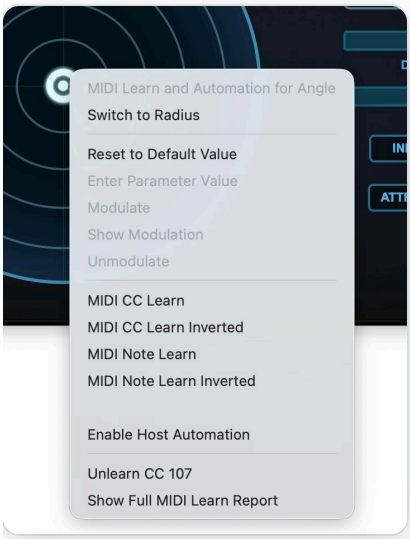


Fig. 11.2 The Orb's menu, set for Angle.

06 **Open the Orb's menu**

Right-click (or Ctrl-click) the Orb.

07 **Make sure it targets Angle**

The first line should read *MIDI Learn and Automation for Angle*, as shown: that is the parameter to set first. If the menu offers *Switch to Angle*, click it and open the menu again.

08 **Start the learn**

Click **MIDI CC Learn**. The menu closes and a flashing box appears around the Orb.

09 **Send the angle**

On the Polar XY device, click **ANGLE** under SEND (MIDI LEARN). The status line confirms what was sent.

10 **Check it took**

The blue box around the Orb stops flashing.

11 **Switch to Radius**

Right-click the Orb and choose **Switch to Radius**.

12 **Start the learn again**

Right-click the Orb again and choose **MIDI CC Learn**. The flashing blue box appears around the Orb.

13 **Send the radius**

On the Polar XY device, click **RADIUS** under SEND (MIDI LEARN).

14 **Save**

Save the Polar XY preset again.

You should now be able to move the Orb with your X/Y controller. If it lands rotated or goes round the wrong way, adjust **Offset** and **Direction** in the MAPPING section (Chapter 6).

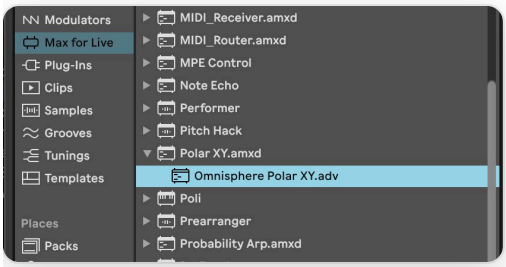


Fig. 11.3 A saved preset under Polar XY.amxd.

That completes the basic setup of the Polar XY Max for Live device with Omnisphere 3. To avoid repeating it each time you use Polar XY, load the preset you created: it appears beneath `Polar XY.amxd` in your Max for Live device listing.

> **TYPING A CC SENDS IT TOO**
> Instead of a SEND button, you can click the Angle or Radius box, type its CC and press Enter during the learn, even if the number is already there. It sends that CC alone, once, just like the button.

When MIDI learn doesn't work

Sometimes Omnisphere 3 refuses to recognise the data coming from the SEND (MIDI LEARN) buttons. There are two things to try.

First, under the MOTION heading in Polar XY, check that **Resolution** is set to *7-bit*, and try the learn steps above again. In 14-bit mode Polar XY sends its MIDI data differently, and this may confuse Omnisphere 3.

If Omnisphere is still stubborn, send the CCs from another source. In Live's browser, under *MIDI Effects*, find the **CC Control** device and drag it between Polar XY and Omnisphere 3:

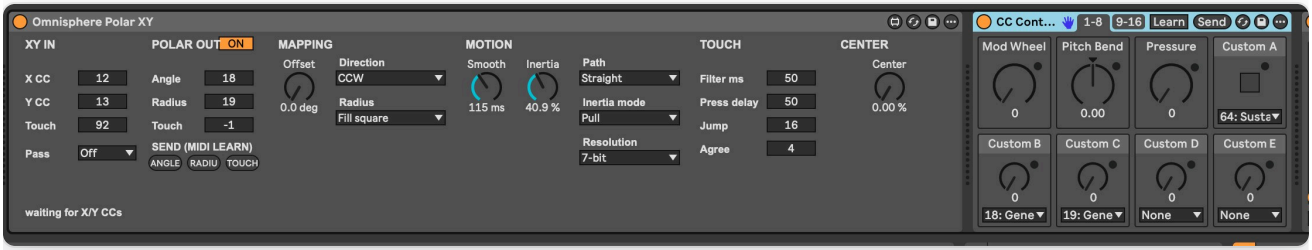


Fig. 11.4 CC Control between Polar XY and Omnisphere, with *Custom B* on CC 18 and *Custom C* on CC 19.

Using the drop-down menus under **Custom B** and **Custom C**, choose the CCs you want to control the Orb with: Custom B for the angle and Custom C for the radius. In the example they are set to CC 18 and CC 19. These must match the **Angle** and **Radius** CCs in Polar XY.

Now repeat the learn steps above, but instead of the SEND (MIDI LEARN) buttons, move the matching knob on the CC Control device: Custom B for the angle, Custom C for the radius. Once both are learned, you can switch off or remove the CC Control device.

CHAPTER 12

Setups and Recipes

The CCs some popular X/Y controllers send out of the box, then starting points for different feels.

Default CCs for select X/Y controllers

These are the CCs each controller sends with its factory settings. Enter the X and Y CCs as Polar XY's **X** and **Y** (**X CC**, **Y CC**), and the touch CC, if there is one, as **TOUCH** under XY IN (**Touch**) for Touch Filter to use (Chapter 8).

CONTROLLER	X CC	Y CC	Z CC	TOUCH CC
Ableton Push 3	74	75	N/A	76
Amptone XY MIDI mini (mode 1)	12	13	N/A	92
Korg NTS-3 Kaoss Pad kit (Total FX)	12	13	14	98
Korg Multi/Poly	18	19	N/A	20
Korg nanoPAD 2	1	2	N/A	16

Polar XY doesn't use a Z CC. It passes through to Omnisphere untouched, where you can MIDI-learn it to something else.

KORG NANOPAD 2
The nanoPAD 2's defaults conflict with other default CCs (its X is on CC 1, the mod wheel), so edit them with the Korg Control Editor before you use it with Polar XY.

WATCH FOR CLASHES
Pick Angle and Radius Out CCs that your controller doesn't send. The Korg Multi/Poly sends X and Y on 18 and 19, which are Polar XY's default Angle and Radius Out, so move the outputs elsewhere, for example to 22 and 23.

Recipes

Load Init first, then set what's listed.

Direct steering

The Orb goes exactly where the controller points, the moment it gets there. Good for joysticks and for checking a new setup.

```
Init · Smoothing 0 ms · Inertia 0%
```

Gentle glide

Takes the edges off quick moves without feeling slow. Most controllers want a little of this.

```
Smoothing 120–250 ms · Inertia 0% · Path Straight
```

Heavy Orb

The Orb lags, swings past where you point and settles back, and drifts on when you let go.

```
Smoothing 100 ms · Inertia 60–80% · Mode Pull
```

Whirlpool

Stir the controller in circles and the Orb keeps spinning after you stop, round the circle rather than across it.

```
Inertia 50% · Path Around · Mode Pull · Center 20%
```

Coasting puck

Only movement matters: flick the controller and the Orb slides off that way and stops wherever it runs out of speed.

```
Inertia 40–60% · Mode Coast · Path Straight
```

Chaos pad

A touch X/Y pad used like a trackpad: drag to push the Orb, let go to fling it off the walls, touch to catch it.

```
Inertia 60% · Mode Pad · Touch In = the pad's touch CC · Touch Filter 40 ms
```

Resistive pad (Geren)

The factory *Geren* preset, tuned on recordings of an Amptone XY MIDI mini, with its default CCs. For another resistive pad, load it and set the CCs to your pad's.

```
X In 12 · Y In 13 · Touch In 92 · Angle / Radius Out 107 / 108 · Touch Filter 50 ms · Jump Limit 24 · Center 15% · Smoothing 150 ms
```

Lift-offs sometimes skitter up to about 50 ms before the touch CC goes off, which is why it holds back 50 ms.

Filter only

Use Polar XY just to clean up a resistive pad's X/Y for some other synth, with no angle or radius at all.

```
Polar Out Off · Pass Input CCs Filtered · Touch Filter 40 ms
```

General tips

- **Learn first, then add motion.** MIDI-learn the Orb with everything at its default, then dial in Smoothing and Inertia.

- **Line up with Offset before you play.** Push the controller hard right and turn Offset until the Orb's dot is on the right.
- **Use the shortest Touch Filter that works.** Every millisecond of it is latency. Try 20 ms, and go up only if the Orb still jumps on touch and release.
- **A touch CC beats guessing.** If your pad sends one, set Touch In: the filter then knows each touch's edges for certain.
- **Automate the feel.** Every setting is a host parameter, so Inertia or Path can change across a song.

APPENDIX A

Parameter Reference

Every parameter Polar XY exposes to the host, with its label on the Max for Live device. Everything is automatable.

PARAMETER	DEVICE LABEL	RANGE	DEFAULT
X In CC	XY IN · X CC	0–127	16
Y In CC	XY IN · Y CC	0–127	17
Touch In CC	XY IN · Touch	Off (-1), 0–127	Off
Pass Input CCs	XY IN · Pass	Off, Raw, Filtered	Off
Polar Out	POLAR OUT · OFF / ON	Off, On	On
Angle Out CC	POLAR OUT · Angle	0–127	18
Radius Out CC	POLAR OUT · Radius	0–127	19
Touch Out CC	POLAR OUT · Touch	Off (-1), 0–127	Off
Angle Offset	MAPPING · Offset	0–360°	0°
Direction	MAPPING · Direction	Counter-clockwise, Clockwise	CCW
Radius Mode	MAPPING · Radius	Clamp to circle, Fill square	Clamp to circle
Smoothing	MOTION · Smooth	0 ms – 2 s	0 ms
Inertia	MOTION · Inertia	0–100%	0%
Motion Path	MOTION · Path	Straight, Around	Straight
Inertia Mode	MOTION · Inertia mode	Pull, Coast, Pad	Pull
Output Resolution	MOTION · Resolution	7-bit, 14-bit	7-bit
Touch Filter	TOUCH · Filter ms	Off (0), up to 100 ms	Off
Press Delay	TOUCH · Press delay	Same (-1), 0–100 ms	Same
Jump Limit	TOUCH · Jump	8–64 steps	16
Agree	TOUCH · Agree	1–16 steps	4
Center	CENTER · Center	Off (0), up to 50%	Off

The MIDI learn buttons (ANGLE, RADIUS, TOUCH, and SEND on the device) are not parameters and can't be automated.

APPENDIX B

Specifications

Formats	CLAP 1.2 note effect (features: note-effect, utility). VST3, wrapped by clap-wrapper, listed as an instrument with a MIDI input and output. Max for Live MIDI effect.
Platforms	macOS universal (10.15+ Intel, 11+ Apple Silicon), Windows (64-bit), Linux (64-bit). The Max for Live device runs wherever Live Suite or Max for Live does.
MIDI	16 channels, each with its own position and motion. Input CCs swallowed (or passed on); everything else passes through.
Conversion	Angle 0–126 (one turn in 127 steps), radius 0–127. Dead zone of about two CC steps around the centre.
Resolution	7-bit, or 14-bit CC pairs (MSB on the CC, LSB on CC + 32) when both output CCs are below 32. 14-bit input read automatically.
Motion rate	One step every 32 samples (plug-in), every 4 ms (Max for Live). Only changed values are sent.
Latency	None, except Touch Filter, which holds input back by its time.
Parameters	21, all automatable.
State	Saved with the host project. User presets are `.pxypreset` files (Chapter 4).
Panel	900 × 738 at 100%; 75%, 100%, 125%, 150% and 200%.

Troubleshooting

SYMPTOM	WHAT TO CHECK
Polar XY doesn't appear in my host (macOS)	The download is probably quarantined. See the warning in Chapter 2, then rescan plug-ins.
The Orb doesn't move	Check that X and Y In match your controller (the XY IN view or the status line should follow it), that Polar Out is *On*, and that Omnisphere learned the Angle and Radius Out CCs. In Live, check the track's MIDI From and Monitor settings.
Omnisphere learned the wrong CC	Learn with the MIDI learn (SEND) buttons rather than moving the controller. If it still won't take, see *When MIDI learn doesn't work* in Chapter 11.
The Orb is rotated or mirrored	Turn **Angle Offset**, or flip **Direction**.
The Orb jumps when I touch or release a pad	A resistive pad. Turn on **Touch Filter** (try 20 to 50 ms), and set Touch In if the pad sends a touch CC.
The Orb spins when the controller rests in the middle	Raise **Center** to 15–25%.
14-bit makes no difference	Both output CCs must be below 32, and the receiving end must read 14-bit pairs.

SYMPTOM	WHAT TO CHECK
Nothing is converted and a warning shows	X and Y In, or Angle and Radius Out, are set to the same CC.
Live: the device's version is old	The status line ends with the version Live loaded. Make sure the newer `Polar XY.amxd` and `polarxy.js` are together in the same folder, and remove older copies.
Live: the VST3 won't go in front of Omnisphere	Live doesn't allow that. Use the Max for Live device (Chapter 10).

Glossary

Cartesian A position given as how far across (X) and how far up (Y).

Polar A position given as a direction (angle) and a distance from the centre (radius).

CC A MIDI continuous controller message: a number 0–127 on one of 128 controller numbers.

MIDI learn Telling a synth to follow whichever CC it receives next.

14-bit CC A pair of CCs, n and n + 32, that together give 16,384 steps instead of 128.

Resistive pad A touch surface that reads position through the finger's contact, which is unreliable as the finger lands and lifts.

Orb Omnisphere 3's circular performance control, which morphs a patch as its dot moves.



Polar XY

TAKE CONTROL OF THE ORB · V0.19.3

github.com/jeffmhopkins/Jclap